Honest vs. Naive Accuracy in HRV-Based Arrhythmia Classification: A Subject-Level Validation Study Using the MIT-BIH Arrhythmia Database

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Abstract:

Cardiac arrhythmias such as premature ventricular contractions (PVCs) can be detected from heart rate variability (HRV), the variation in timing between successive heartbeats. Five recordings from the MIT-BIH Arrhythmia Database were analyzed in R, using expert cardiologist beat annotations to build RR intervals directly rather than detecting the beats again. HRV features (SDNN, RMSSD, pNN50, CVrr) were computed over sliding 30-beat windows and used to train a Random Forest classifier.

The central question was whether the train/test split changes reported accuracy. A naive split, mixing windows from the same patient across both sides, scored 97.3%. An honest subject-level split, holding out entire patients, scored 79.6%. Naive validation substantially overstates real-world performance.

Background Information:

- Cardiac arrhythmias, including PVCs, can occur without obvious external symptoms, making automated detection valuable for patient monitoring.
- Heart rate variability (HRV), the beat-to-beat variation in timing between heartbeats, changes measurably during arrhythmic events.
- Many published arrhythmia classifiers report very high accuracy, but some of that accuracy comes from data leakage: allowing beats from the same patient into both the training and test sets.
- A subject-level (inter-patient) split, where no patient's beats appear on both sides, gives a realistic estimate of performance on a new, unseen patient.

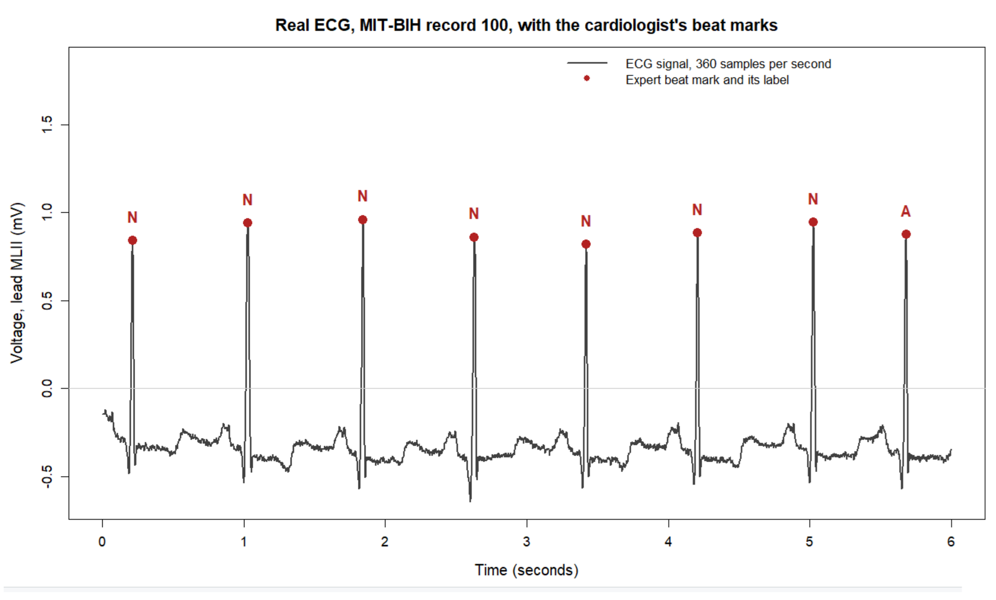


Figure 1: Raw lead-MLII ECG from record 100 with the cardiologist's beat marks overlaid. These expert beat positions are the ground truth used to build every RR interval in this study.

Method/Materials:

- Data: Five real recordings from the MIT-BIH Arrhythmia Database (records 100, 103, 106, 119, 208), sampled at 360 Hz, each paired with expert cardiologist beat annotations.
- Beat extraction: Used the expert-labeled beat positions directly from the annotation files rather than a custom R-peak detector, preserving the accuracy of the ground truth.
- HRV computation: RR intervals were computed between consecutive expert-labeled beats. Features (meanRR, meanHR, SDNN, RMSSD, pNN50, CVrr) were calculated over sliding windows of 30 beats.
- Classification: A Random Forest classifier (400 trees) was trained to separate normal-rhythm windows from arrhythmia windows (windows containing at least one PVC).
- Validation: Records 100, 106 and 119 were used for training; records 103 and 208 were held out entirely for testing. A naive random-window split was also run for comparison.
- Software: RStudio with the randomForest package and base R.

Results:

- Subject-level (honest) split: 79.6% accuracy, 100% specificity, 64.6% sensitivity on 167 held-out test windows.
- Naive random-window split: 97.3% accuracy on the same data, a 17.7-point overstatement.
- PVCs show a wider, deeper averaged waveform and arrive earlier than normal beats (Fig. 2).
- The PVC-heavy record shows short-long RR jumps absent from the normal-heavy record (Fig. 3).
- RMSSD (33.8) and SDNN (29.7) ranked as the most important features; meanRR (2.4) and meanHR (1.6) contributed least (Fig. 5).

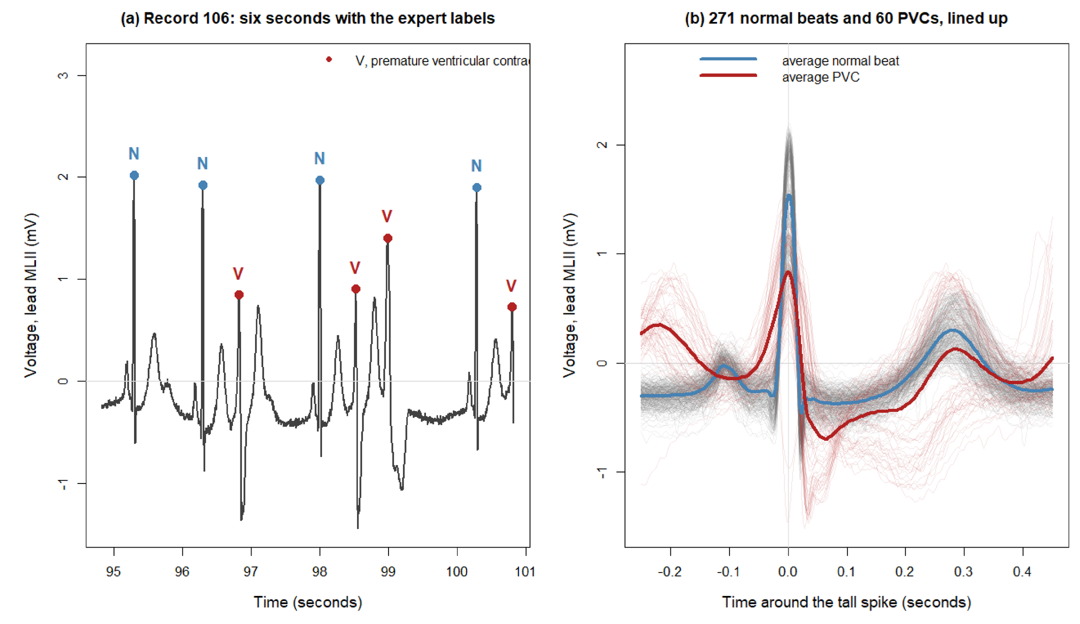


Figure 2: Record 106: six seconds with expert labels (left), and 271 normal beats against 60 PVCs lined up on their tall spike (right). The average PVC is wider and dips deeper than the average normal beat.

Figure 3: RR interval series (tachograms) for normal-heavy record 100 (left) and PVC-heavy record 208 (right). Record 208 shows the short-long jumps caused by an early PVC followed by a compensatory pause.

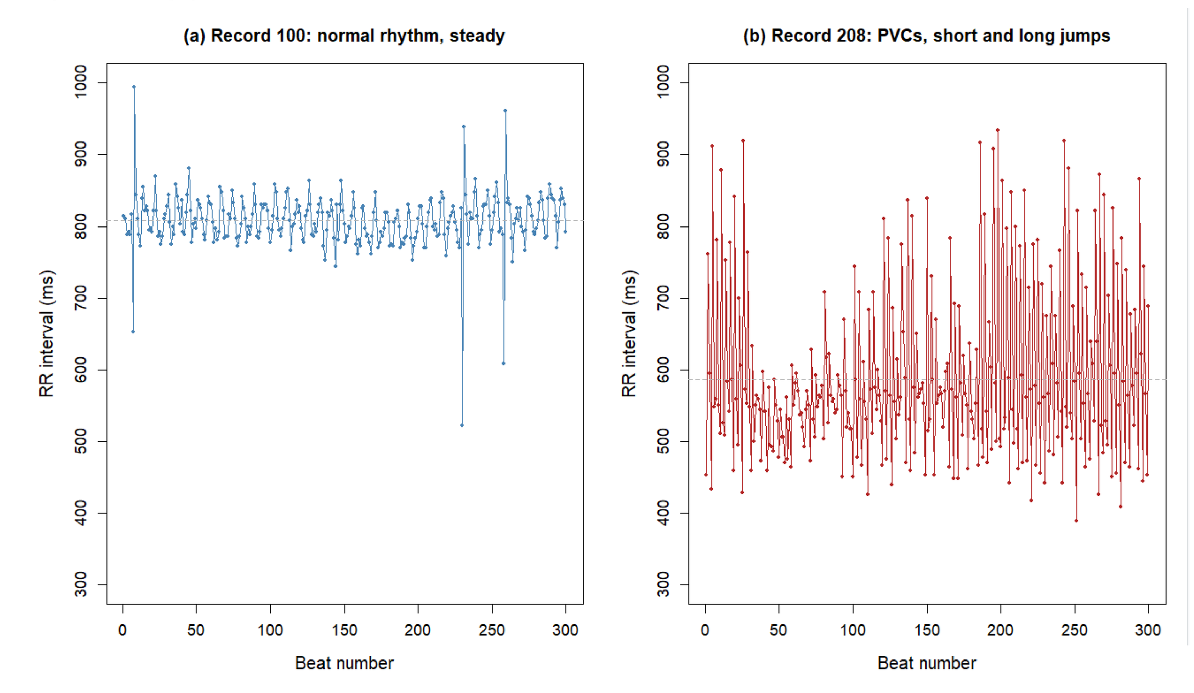


Figure 4: Confusion matrix under the subject-level split (trained on 100, 106, 119; tested on held-out 103 and 208): 79.6% accuracy, 100% specificity, 64.6% sensitivity. A naive random-window split scores 97.3% on the same data.

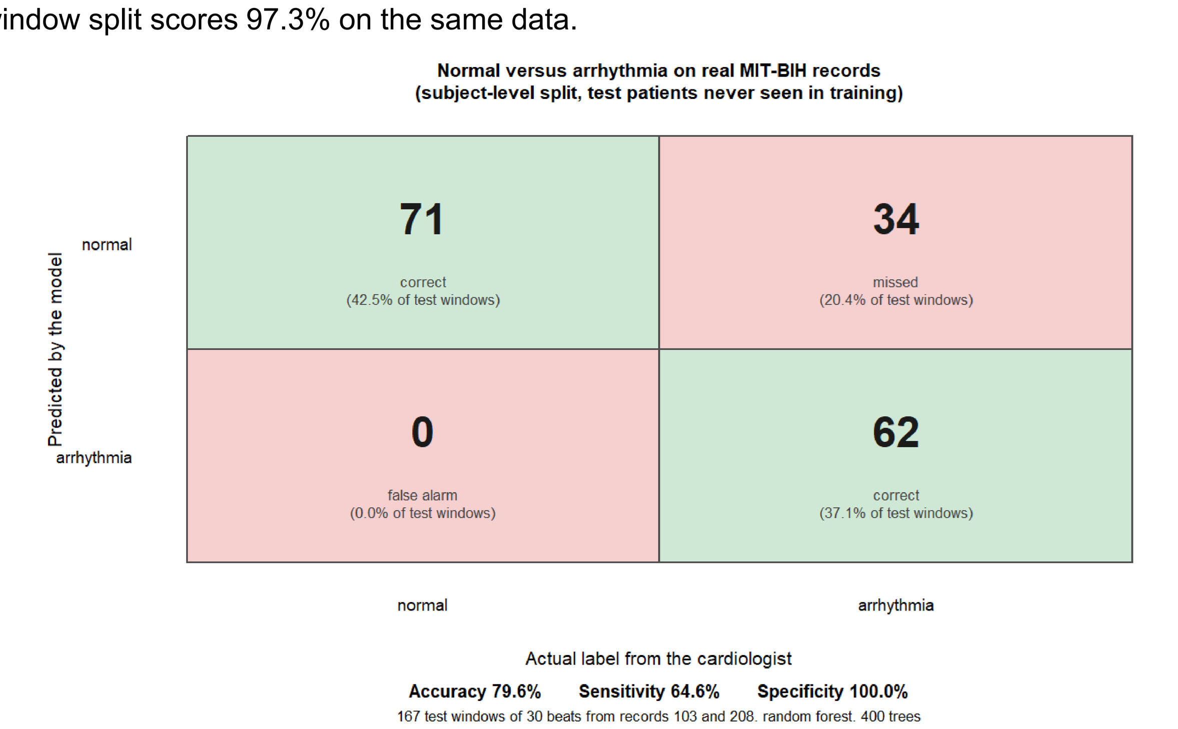
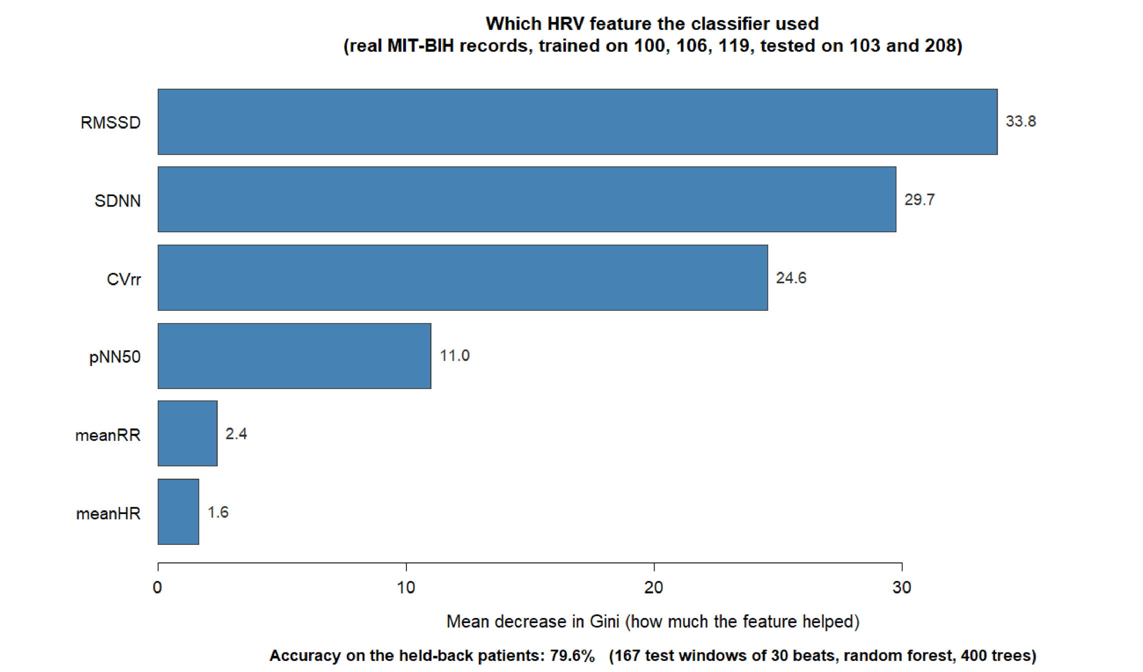


Figure 5: Importance of each HRV feature to the classifier. RMSSD and SDNN, both measures of beat-to-beat irregularity, rank highest, well ahead of mean RR interval and mean heart rate.



Discussion/Conclusion:

- HRV features from expert-annotated RR intervals, with a Random Forest classifier, separate normal rhythm from PVC-containing rhythm at 79.6% accuracy under an honest, subject-level split.
- The 17.7-point gap between honest (79.6%) and naive (97.3%) accuracy shows that validation scheme matters as much as model choice.
- Perfect specificity (100%) with moderate sensitivity (64.6%) means the model is conservative: it rarely raises false alarms but misses some true arrhythmia windows.
- That irregularity features (RMSSD, SDNN) outrank rate features (meanRR, meanHR) indicates the classifier keys on beat-timing irregularity rather than heart rate itself.
- Future work: scaling to more of the 48 MIT-BIH records, comparing Random Forest to an artificial neural network (ANN), and building an R Shiny app for real-time classification.

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